

Engineering Formula Handbook

Part 1: Core Formulas

1. General mathematics and physics

These formulas are used in nearly every engineering branch.

Motion

- Speed: $v = \frac{d}{t}$
- Velocity: $v = \frac{\Delta x}{\Delta t}$
- Acceleration: $a = \frac{v-u}{t}$
- Force: $F = ma$
- Momentum: $p = mv$
- Impulse: $J = F\Delta t = \Delta p$

Work, energy, power

- Work: $W = Fs$
- Power: $P = \frac{W}{t}$
- Kinetic energy: $KE = \frac{1}{2}mv^2$
- Potential energy: $PE = mgh$
- Efficiency: $\eta = \frac{\text{output}}{\text{input}} \times 100\%$

Density and pressure

- Density: $\rho = \frac{m}{V}$
- Pressure: $P = \frac{F}{A}$
- Stress: $\sigma = \frac{F}{A}$
- Strain: $\varepsilon = \frac{\Delta L}{L}$

Trigonometry basics

- $\sin^2 \theta + \cos^2 \theta = 1$
- $\tan \theta = \frac{\sin \theta}{\cos \theta}$
- $\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$
- $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

2. Electrical engineering formulas

Electrical formula sets usually begin with circuit laws, then move into AC, power, magnetics, machines, and power systems.

Basic circuit laws

- Ohm's law: $V = IR$
- Power: $P = VI = I^2 R = \frac{V^2}{R}$
- Electrical energy: $E = Pt$
- Resistance in series: $R_{eq} = R_1 + R_2 + \dots$
- Resistance in parallel: $\frac{1}{R_{eq}} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$

AC quantities

- RMS voltage: $V_{rms} = \frac{V_m}{\sqrt{2}}$
- RMS current: $I_{rms} = \frac{I_m}{\sqrt{2}}$
- Frequency: $f = \frac{1}{T}$
- Angular frequency: $\omega = 2\pi f$

Capacitors and inductors

- Capacitor charge: $Q = CV$
- Capacitor energy: $E = \frac{1}{2} CV^2$
- Inductor voltage: $V = L \frac{di}{dt}$
- Inductor energy: $E = \frac{1}{2} LI^2$

Magnetic circuits

- Magnetic flux: $\Phi = BA$
- Flux density: $B = \frac{\Phi}{A}$
- Magnetomotive force: $F_m = NI$
- Reluctance: $\mathcal{R} = \frac{l}{\mu A}$

Three-phase power

- Apparent power: $S = VI$
- Real power: $P = VI \cos \phi$
- Reactive power: $Q = VI \sin \phi$
- Three-phase power: $P = \sqrt{3} V_L I_L \cos \phi$

3. Electronics formulas

This section is useful for diodes, transistors, op-amps, filters, and signal circuits.

Diodes and semiconductors

- Diode equation: $I = I_S \left(e^{\frac{V_D}{nV_T}} - 1 \right)$
- Thermal voltage: $V_T = \frac{kT}{q}$
- RC time constant: $\tau = RC$

Transistors

- BJT current relation: $I_E = I_B + I_C$
- Current gain: $\beta = \frac{I_C}{I_B}$
- MOSFET drain current depends on region of operation and gate-source voltage.

Op-amp formulas

- Inverting amplifier: $V_o = -\frac{R_f}{R_{in}} V_i$
- Non-inverting amplifier: $V_o = \left(1 + \frac{R_f}{R_1} \right) V_i$
- Summing amplifier output is proportional to the weighted sum of inputs.

Signal processing basics

- Gain in dB: $G_{dB} = 20\log_{10}\left(\frac{V_{out}}{V_{in}}\right)$
- Power gain in dB: $G_{dB} = 10\log_{10}\left(\frac{P_{out}}{P_{in}}\right)$

4. Mechanical engineering formulas

Mechanical handbooks usually cover strength of materials, statics, dynamics, thermodynamics, fluids, and machine design.

Statics and strength of materials

- Stress: $\sigma = \frac{F}{A}$
- Strain: $\varepsilon = \frac{\Delta L}{L}$
- Young's modulus: $E = \frac{\sigma}{\varepsilon}$
- Shear stress: $\tau = \frac{F}{A}$
- Torque: $\tau = Fr$

Rotation and dynamics

- Angular velocity: $\omega = \frac{\theta}{t}$
- Tangential velocity: $v = r\omega$
- Centripetal force: $F = \frac{mv^2}{r}$

Fluids

- Continuity equation: $A_1V_1 = A_2V_2$
- Bernoulli's equation: $P + \frac{1}{2}\rho v^2 + \rho gh = \text{constant}$
- Flow rate: $Q = AV$

Thermodynamics

- Heat transfer: $Q = mc\Delta T$
- First law: $\Delta U = Q - W$
- Thermal efficiency: $\eta = \frac{W_{out}}{Q_{in}}$

5. Civil engineering formulas

Civil engineering formula sets commonly include surveying, structural analysis, hydraulics, soil mechanics, and concrete design.

Structural formulas

- Stress: $\sigma = \frac{F}{A}$
- Bending stress: $\sigma = \frac{My}{I}$
- Section modulus: $Z = \frac{I}{y}$
- Shear stress: $\tau = \frac{VQ}{Ib}$

Surveying and geometry

- Area of triangle: $A = \frac{1}{2}bh$
- Volume of cuboid: $V = lbh$
- Slope: $m = \frac{\Delta y}{\Delta x}$

Fluid and hydraulic formulas

- Discharge: $Q = AV$
- Pressure head: $h = \frac{P}{\rho g}$
- Manning's equation is used for open-channel flow.

Concrete and materials

- Compressive strength is measured by load divided by area.
- Modulus of elasticity and safety factor are key design values.

6. Chemical and materials formulas

A good engineering handbook should also include basic chemical and materials formulas because many industrial problems involve them.

Core formulas

- Moles: $n = \frac{m}{M}$
- Concentration: $C = \frac{n}{V}$
- Density: $\rho = \frac{m}{V}$
- Percentage yield: $\% \text{yield} = \frac{\text{actual}}{\text{theoretical}} \times 100\%$

Materials relations

- Stress: $\sigma = \frac{F}{A}$
- Strain: $\varepsilon = \frac{\Delta L}{L}$
- Hooke's law: $\sigma = E\varepsilon$

7. Useful constants

These should appear in the formula handbook so users do not need to search elsewhere.

- Gravitational acceleration: $g \approx 9.81 \text{ m/s}^2$
- Speed of light: $c \approx 3 \times 10^8 \text{ m/s}$
- Universal gas constant: $R = 8.314 \text{ J/(mol} \cdot \text{K)}$
- Planck constant: $h \approx 6.626 \times 10^{-34} \text{ J} \cdot \text{s}$
- Electron charge: $q \approx 1.602 \times 10^{-19} \text{ C}$

8. Formula usage tips

Always check whether the formula requires SI units, because many formulas break if units are mixed incorrectly. Engineering conversion references emphasize consistent unit systems, significant figures, and dimensional checking before substitution.

Engineering Unit Conversion Handbook

1. SI prefixes

SI prefixes are used to express very large or very small values in a compact way. This is one of the first things every engineering student should memorize.

Prefix	Symbol	Factor
tera	T	10^{12}
giga	G	10^9
mega	M	10^6
kilo	k	10^3
hecto	h	10^2
deka	da	10^1
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}

Quick examples

- $1 \text{ km} = 1000 \text{ m}$
- $1 \text{ mA} = 10^{-3} \text{ A}$
- $1 \text{ }\mu\text{F} = 10^{-6} \text{ F}$
- $1 \text{ MPa} = 10^6 \text{ Pa}$

2. Length conversions

Length conversions are among the most frequently used in engineering problems.

Unit	Equivalent
1 mm	10^{-3} m
1 cm	10^{-2} m
1 m	100 cm
1 km	1000 m
1 inch	25.4 mm
1 foot	12 inches
1 yard	3 feet
1 mile	5280 feet

Useful relation

- $1 \text{ in} = 2.54 \text{ cm}$
- $1 \text{ ft} = 0.3048 \text{ m}$

3. Area and volume conversions

Area and volume often require careful squaring or cubing of conversion factors.

Unit	Equivalent
1 cm^2	10^{-4} m^2
1 m^2	10^4 cm^2
1 cm^3	10^{-6} m^3
1 L	10^{-3} m^3
1 m^3	1000 L
1 ft^3	28.3168 L

Important reminder

Always square or cube the length conversion factor for area and volume conversions.

4. Mass and force conversions

These are especially important in mechanics, statics, and structural calculations.

Unit	Equivalent
1 g	10^{-3} kg
1 kg	1000 g
1 metric ton	1000 kg
1 N	$1 \text{ kg} \cdot \text{m/s}^2$
1 kN	1000 N
1 lb_f	4.44822 N
1 lbf	4.44822 N

5. Pressure conversions

Pressure units appear in fluids, hydraulics, thermodynamics, and structural work.

Unit	Equivalent
1 Pa	1 N/m ²
1 kPa	1000 Pa
1 MPa	10 ⁶ Pa
1 bar	10 ⁵ Pa
1 atm	101325 Pa
1 psi	6894.76 Pa

Common engineering note

MPa is often used for stress, while bar and atm are common in fluid systems.

6. Energy and power conversions

Energy and power are central in electrical, mechanical, and thermal engineering.

Unit	Equivalent
1 J	1 N·m
1 kJ	1000 J
1 Wh	3600 J
1 kWh	3.6 MJ
1 cal	4.184 J
1 hp	746 W

7. Temperature conversions

Temperature conversion is simple in formula form but easy to misuse if the scale is not identified clearly.

Conversion	Formula
Celsius to Kelvin	$K = C + 273.15$
Kelvin to Celsius	$C = K - 273.15$
Celsius to Fahrenheit	$F = \frac{9}{5}C + 32$
Fahrenheit to Celsius	$C = \frac{5}{9}(F - 32)$

8. Time conversions

Time conversions are basic, but they appear constantly in physics, scheduling, and process engineering.

Unit	Equivalent
1 min	60 s
1 hr	60 min
1 day	24 hr
1 week	7 days
1 year	365 days approximately

9. Electrical and electronic unit conversions

These conversions are especially important in circuits, semiconductor devices, and signal systems.

Unit	Equivalent
1 V	1 J/C
1 A	1 C/s
1 Ω	1 V/A
1 F	1 C/V
1 H	1 V·s/A
1 μF	10^{-6} F
1 mA	10^{-3} A
1 k Ω	1000 Ω

Signal units

- 1 dB = $10\log_{10}(P_2/P_1)$ for power ratio.
- $20\log_{10}(V_2/V_1)$ for voltage ratio.

10. Fluid and flow units

These units are common in civil, mechanical, chemical, and process engineering.

Unit	Equivalent
1 m ³ /s	1000 L/s
1 L/s	10 ⁻³ m ³ /s
1 gal (US)	3.78541 L
1 ft ³ /s	0.0283168 m ³ /s
1 m/s	3.6 km/h

11. Angular and rotational units

These are used in mechanics, robotics, machinery, and control systems.

Unit	Equivalent
1 revolution	360°
1 revolution	2 π rad
1 rad	57.2958°
1°	0.0174533 rad

12. Conversion method

A reliable conversion method is to multiply by a conversion factor arranged so unwanted units cancel. This is often called dimensional analysis or factor-label method.

Example

Convert 5 km to meters:

$$5 \text{ km} \times \frac{1000 \text{ m}}{1 \text{ km}} = 5000 \text{ m}$$

Another example

Convert 2.5 MPa to Pa:

$$2.5 \text{ MPa} \times \frac{10^6 \text{ Pa}}{1 \text{ MPa}} = 2.5 \times 10^6 \text{ Pa}$$

13. Common conversion mistakes

Engineering references often warn against these mistakes:

- Mixing mm and m without conversion.
- Forgetting to square or cube the factor for area or volume.
- Using Celsius directly in formulas requiring Kelvin.
- Confusing weight and mass.
- Using wrong pressure units in stress problems.

14. Quick reference summary

Must-remember conversions

- 1 in = 25.4 mm
- 1 ft = 0.3048 m
- 1 kN = 1000 N
- 1 MPa = 10^6 Pa
- 1 kWh = 3.6×10^6 J
- $K = C + 273.15$
- 1 rev = 2π rad

Engineering Symbols & Standards Guide

1. Why standards matter

Engineering standards make drawings readable across teams, companies, and countries. They reduce mistakes in manufacturing, inspection, maintenance, and installation because everyone uses the same conventions for lines, symbols, dimensions, and notes.

Main benefits

- Clear communication.
- Fewer design errors.
- Better manufacturability.
- Faster review and approval.
- Consistent documentation.

2. Major standards bodies

ISO

ISO publishes standards for technical drawings, graphical symbols, product specifications, and component representation used globally in engineering and manufacturing.

ASME

ASME Y14 standards define many drawing practices, including engineering drawing rules, dimensioning, and product definition conventions.

IEC

IEC standards support graphical symbols used on electrical and electronic equipment and ensure consistency in product publications.

ISA

ISA conventions are widely used in instrumentation and process diagrams, especially P&ID notation.

Local standards

National standards, such as BIS/IS in India or BS in the UK, may adapt or adopt international conventions for local use.

3. Drawing line types

Line types are one of the most basic parts of engineering graphics. They tell the viewer what is visible, hidden, cut, cantered, or dimensioned.

Line type	Meaning
Continuous thick	Visible outlines and edges.
Continuous thin	Dimensions, extension lines, hatching, leader lines.
Dashed thin	Hidden edges and outlines.
Chain thin	Center lines, axes, symmetry.
Chain thick or special line	Cutting plane or special reference lines.

4. Dimensioning symbols

Dimensioning symbols are used to describe size and tolerance on drawings. They are essential in mechanical and manufacturing drawings.

Common symbols

- Diameter: $\varnothing$
- Radius: R
- Square: $\square$
- Depth: D or depth symbol depending on standard
- Counterbore.
- Countersink.
- Tolerance plus/minus: $\pm$

Example

- $\varnothing 25$ means a 25 mm diameter hole.
- R10 means a 10 mm radius.
- 20 ± 0.1 means acceptable size range around 20 mm.

5. GD&T basics

Geometric Dimensioning and Tolerancing, or GD&T, is a symbolic language used to define the form, orientation, location, and profile of features. It helps control how parts fit and function.

Main GD&T categories

- Form.
- Orientation.
- Location.
- Profile.
- Runout.

Why it matters

GD&T reduces ambiguity and allows functional tolerances instead of only size tolerances.

6. Electrical symbols

Electrical and electronic drawings use standard symbols for circuits, devices, and components. IEC and related references are used to keep equipment diagrams understandable.

Common symbols

- Resistor.
- Capacitor.
- Inductor.
- Diode.
- LED.
- Transistor.
- Switch.
- Fuse.
- Battery.
- Ground.

Reading rule

A symbol may look slightly different by drawing standard, but the function should remain clear and consistent across the schematic.

7. P&ID and process symbols

Piping and instrumentation diagrams use standardized symbols for valves, pumps, tanks, instruments, and flow paths. These diagrams are important in chemical, mechanical, oil-and-gas, and process engineering.

Common symbols

- Valve symbols.
- Pump symbols.
- Tank/vessel symbols.
- Flow arrows.
- Instrument bubbles.
- Pipe lines and fittings.

Instrument tags

Tag letters and bubbles are used to identify what an instrument measures or controls. ISA-style conventions are common in process diagrams.

8. Mechanical and piping symbols

Mechanical drawings use symbols for fittings, welds, threads, and assembly features. Isometric and fabrication drawings often require specialized symbol reading.

Examples

- Elbow 90°.
- Elbow 45°.
- Tee equal.
- Reducer concentric.
- Cap.
- Butt weld symbol.
- Socket weld symbol.
- Threaded symbol.

Practical use

These symbols help fabrication teams know exactly what part is needed and how it should be connected.

9. Civil engineering symbols

Civil and architectural drawings use symbols for walls, doors, windows, levels, sections, structural members, and site features. ISO-style graphical conventions help keep plans and construction drawings consistent.

Common symbols

- Door swings.
- Window markers.
- Section arrows.
- Grid lines.
- Level indicators.
- Section cuts.
- Rebar and structural notation.

10. Common abbreviations

Engineering drawings often use short abbreviations to save space. These must be used carefully and consistently.

Abbreviation	Meaning
TYP	Typical
NTS	Not to scale
REF	Reference
DIA	Diameter
R	Radius
CL	Center line
CBORE	Counterbore
CSK	Countersink
EQ	Equal
MIN	Minimum
MAX	Maximum

11. Surface finish and machining symbols

Surface texture symbols are used to specify how smooth or rough a surface must be after manufacturing. Machining symbols help communicate if a surface should be left as-made or finished.

Common meanings

- Basic surface texture symbol.
- Machining required.
- Material removal not permitted.
- Direction of lay.
- Roughness value.

Why it matters

Surface finish affects fit, friction, sealing, appearance, and performance.

12. Welding symbols

Welding symbols are used on fabrication drawings to show the type, location, size, and shape of welds. They are essential in structural, piping, and mechanical fabrication.

Common parts

- Arrow.
- Reference line.
- Tail.
- Weld symbol.
- Size notation.
- Finish symbol.

Weld types

- Fillet weld.
- Groove weld.
- Plug weld.
- Spot weld.
- Seam weld.